

Yuewen Gao

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RESEARCH INTERESTS

Human-Agent Collaboration: Designing mixed-initiative interactions and adaptive LLM agents that infer user intent, foster mutual mental models, and preserve human agency grounded in cognitive theories.

Information Visualization & Intelligent Exploration: Creating interactive visualization systems that integrate analytical provenance and structure-aware guidance to facilitate progressive knowledge discovery across complex data.

AI-Augmented Creative & Knowledge Work: Developing proactive AI-assisted tools and interaction paradigms to enhance human ideation, dynamic reasoning, and decision-making in complex, open-ended tasks.

EDUCATION

2023 - Present **Nanjing University**
B.S. in Intelligent Science and Technology (*expected 2027*) · GPA: 4.44/5.00

RESEARCH EXPERIENCE

2026 -- Present **Northeastern Human-Centered AI Lab** *Advised by Bingsheng (Arthur) Yao*

- **Human-Agent Intent Co-Construction** (Ongoing):
 - Performed hybrid coding (deductive and inductive) of human behaviors and self-reported action-level intent in collaboration experiments (Map Task).
 - Derived a design framework for AI agents to establish, externalize, and repair grounding during evolving human-agent collaboration
 - Led agent prototyping, harness iterative refinement, and human-subject experiments.
- **Human-LLM Interaction Trajectory in Team Deliberation** (Under Review at CHI 2027) [M.2]:
 - Conducted qualitative coding of human team behavioral logs from a 16-team study to map team collaboration patterns for investigating how shareable and continuable human-LLM interaction trajectories shape team deliberation
- **Trajectory-Centered Multi-Agent Coordination** (Under Review at CHI 2027) [M.3]:
 - Conducted semi-structured interviews with 10 participants and inductive coding to examine how trajectory-centered paradigms reshape knowledge workers' cognitive workflows across multi-agent coordination
 - Drafted manuscript for the qualitative findings, contextualizing empirical insights with representative participant accounts.

2025 -- Present **iMate Lab, Nanjing University** *Advised by Prof. Ke Xu*

- **Non-intrusive Assistance for Sketch-Based Ideation** (Under Review at CHI 2027) [M.1]:
 - Conducted statistical and thematic analysis in an 11-participant formative study to map distinctive drawing behaviors to creative support needs.
 - Derived a theory-grounded framework that orchestrates whether, when, and what support to surface based on behavioral evidence and canvas context.
 - Led end-to-end system prototyping and validated efficacy through a 14-participant controlled comparison and a 24-participant component-ablation study.

- 2025 -- Present
- **Multimodal Visualization for News Video Sensemaking** (Under Review at Visual Informatics) [M.4]:
 - Architected an unsupervised multimodal pipeline combining vision-textual features for event boundary detection and cross-broadcast semantic association.
 - Led the full-stack engineering of *Newstract*, building an interactive visual analytics system with dynamic visualization components for progressive exploration.
 - Combined quantitative algorithm benchmarking, controlled user studies, and domain case studies to evaluate system performance, user efficacy, and analytical workflows.
 - **Adaptive Recommendations for Network Exploration** (ACM CHI 2026) [C.1]:
 - Designed and implemented the Analytic State Graph provenance module, enabling branching history tracking and context-aware hypothesis exploration.
 - Developed full-stack interactive components (React, D3.js, G6.js) integrating user interaction feedback with the adaptive recommendation engine.
 - Contributed to manuscript drafting and quantitative data analysis across controlled user evaluations comparing baseline exploration.

PUBLICATIONS & MANUSCRIPTS

Manuscripts Under Review

- CHI 2027
- [M.1] YUEWEN GAO, Wenchao Li, Yiqing Yang, Lejia Li, Yutong Wu, Ke Xu. “ProactiveFlow: Coordinating Proactive AI Support for Sketch-Based Ideation”. Under Review at CHI 2027
- [M.2] Jiaju Chen, Jingfei Huang, Chaoran Chen, YUEWEN GAO, Yun Wang, April Yi Wang, Toby Jia-Jun Li, Jian Zhao, Dakuo Wang, Bingsheng Yao. “Carrying Thoughts Forward: Exploring How Shareable and Continuable Human-LLM Conversation Trajectories Reshape Team Deliberation”. Under Review at CHI 2027
- [M.3] Menglin Zhao, Fangrui Sima, Yuxuan Lu, YUEWEN GAO, Chaoran Chen, Xuhai “Orson” Xu, Jian Zhao, Toby Jia-Jun Li, Dakuo Wang, Bingsheng Yao. “Trajectory-Centered Interaction: User-Directed Coordination of Multiple AI Agents in Knowledge Work”. Under Review at CHI 2027
- Visual Informatics
- [M.4] YUEWEN GAO, Yu He, Xianglei Lv, Yingying Dong, Ke Xu. “Multimodal Analysis of News Videos: Event-Based Visual Summarization over Serial Broadcasts”. Under Review at Visual Informatics

Conference Publications

- CHI 2026
- [C.1] Wenchao Li, YUEWEN GAO, Yu He, Cong Zhu, Ke Xu 2026. “NetworkCanvas: Supporting Progressive Network Visualization Exploration via Adaptive Recommendations”. In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 410:1–410:22. DOI: [10.1145/3772318.3791211](https://doi.org/10.1145/3772318.3791211)

AWARDS & HONORS

- Nov 2025
- People’s Scholarship**
University-level Merit Scholarship
- Jan 2025
- Honorable Mention**
Mathematical Contest in Modeling (MCM)
- Nov 2024
- People’s Scholarship**
University-level Merit Scholarship

SKILLS

Programming Languages: Python, TypeScript, JavaScript, C/C++

Libraries & Platforms: React, Node.js, PyTorch, D3.js, G6.js, ECharts, Git, Docker

Research: Quantitative & Qualitative methods, System Development, Experiment Design

Languages: Chinese (Native), English (TOEFL iBT 105, CET-4 658, CET-6 640)